

ANSWERS

1.

Question	Answer	Marks
8(a)	steel	B1
8(b)(i)	magnetic field (lines of magnet) cut (by solenoid) or magnetic field (in solenoid) changes	B1
	electromotive force / e.m.f. <u>induced</u> / current <u>induced</u>	B1
	electromotive force / e.m.f. cause current (in closed circuit)	B1
8(b)(ii)	Lenz law mentioned or to oppose the change (causing the current)	B1
	(In order to) repel magnet (upwards) or upward force on magnet or repulsion (of magnet) mentioned or to oppose the magnet or resists motion of magnet	B1
8(c)	current / it changes direction	B1
	magnitude / size of current changes or current increases / decreases	B1

2.

Question	Answer	Marks
7(a)(i)	P: (carbon) brush and Q: slip ring(s)	B1
7(a)(ii)	to connect the output terminal(s) and the (rotating) coil/brushes	B1
	without wires twisting	B1
7(b)(i)	at least one of the five positions when the output e.m.f. is zero indicated with X and no indications elsewhere	B1
7(b)(ii)	(amplitude $\equiv$) 3.6 (cm) or ($E_{\max}$ $\equiv$) 5.0×3.6 or 36 (V) seen	C1
	18 (V)	A1
7(b)(iii)	(T $\equiv$) 8.0 (m s)	B1
7(b)(iv)	(f $\equiv$) $1/T$ or $1/8.0$ or $1/(4.0 \times 2.0)$ or $1/0.0080$	C1
	130 (Hz)	A1

3.

Question	Answer	Marks
8(a)	(soft) iron	B1
8(b)	input voltage or input current is alternating	B1
	changing (magnetic) field (in primary causes changing magnetic field in secondary) (alternating emf is) <u>induced</u> (in secondary coil)	B1
	or (magnetic) field lines cut secondary	B1
8(c)	(N_s $\equiv$) $4600 \times 5 / 230$ or $V_p / V_s = N_p / N_s$ in any form	C1
	100	A1
8(d)	reverse the coils OR connect the input to the secondary / right-side coil OR secondary has more turns than <u>primary</u> / more than 4600 (with same number on primary)	B1

4.

Question	Answer	Marks
9(a)	<u>magnetic</u> field or <u>magnetic</u> flux mentioned (from magnets)	B1
	it / sides of <u>coil</u> cut (magnetic) field or (magnetic) field (strength) / flux in <u>coil</u> changes	B1
9(b)	AB on right – 6 V	B1
	coil vertical 0 V in both positions	B1
9(c)(i)	one ring (instead of two) with gaps between the two halves	C1
	each side of split ring connected to a different end of the coil and to a brush	A1
9(c)(ii)	(force on) current in a magnetic field or left-hand rule mentioned or magnetic field of coil interacts with field of magnet	B1

5.

Question	Answer	Marks
6(a)	<u>magnetic</u> field / flux mentioned	B1
	changing / alternating (magnetic) field	B1
	voltage <u>induced</u> (in secondary coil)	B1
6(b)(i)	$(V_S =) V_P \times N_S / N_P$ or $240 \times 350 / 5600$	C1
	15 V	A1
6(b)(ii)	$(I =) P / V$ or $90 / 15$	C1
	6.0 A	A1

6.

Question	Answer	Marks
6(a)	soft iron	B1
6(b)(i)	$(P =) VI$ or 220×39	C1
	8.6×10^3 W	A1
6(b)(ii)	$(Q =) It$ or 39×4.5 or $39 \times 4.5 \times 60$ or $39 \times 4.5 \times 60 \times 60$ or $39 \times 4.5 \times 3600$	C1
	180 or 11 000 or $39 \times 4.5 \times 60 \times 60$ or $39 \times 4.5 \times 3600$	C1
	6.3×10^5 C	A1
6(c)	resistance increases with temperature or resistance lower (because of (low / ambient) temperature)	B1

7.

Question	Answer	Marks
9(a)	always positive or never negative or direction does not change	B1
	constant value or size does not change	B1
9(b)(i)	$(I =) V / R$ or $12 / (20 + 28)$ or $12 / 48$	C1
	0.25 A	A1
9(b)(ii)	YZ in magnetic <u>field</u> or magnetic <u>field</u> between poles (of magnet)	M1
	(electric) current in YZ or interacts with field of current / YZ	A1
9(b)(iii)	downwards; middle, right-hand box ticked i.e. towards N ☐ towards S ☐ towards Y ☐ towards Z ☐ downwards ☒ upwards ☐	B1
9(b)(iv)	current from Z to Y page / into page and magnetic field left to right	B1
	(Fleming's) left-hand rule / other appropriate rule mentioned / described	B1
9(c)(i)	$1 / R_T = 1 / R_1 + 1 / R_2$ or $(R_T =) R_1 R_2 / (R_1 + R_2)$ or $1 / R_T = 1 / 20 + 1 / 30$ or $1 / R_T = 5 / 60$ or $1 / R_T = 1 / 12$ or $(R_T =) 20 \times 30 / (20 + 30)$	C1
	12	C1
	40 W	A1
9(c)(ii)	resistance decreases or resistance is less (with S_2 closed)	B1
	current increases and force increases	B1
9(c)(iii)	1.5	B1

8.

6(a)(i)	top magnet N pole at bottom and S pole at top	B1
	like poles repel or N and N repel	B1
6(a)(ii)	iron magnetised (by induction) / iron a temporary magnet / poles produced on iron (by magnet)	B1
	unlike poles attract or N -pole attracts S-pole or bottom pole of iron is a S pole or S pole (on iron) faces N-pole (on magnet)	B1
6(b)	at least three circles centered on wire	B1
	clockwise direction marked on at least one closed shape around wire and no arrows wrong	B1

9.

Question	Answer	Marks
5(a)	brush / carbon	B1
	split-ring or commutator	B1
5(b)(i)	faster or greater turning effect	B1
5(b)(ii)	goes backwards / turns backwards / reverses	B1
5(c)(i)	speeds up and slows down (when turned on / off)	B1
	reaches a steady speed / terminal speed / maximum speed or slows down but never to rest or cycle continuously repeats	B1
5(c)(ii)	same top / constant speed or constant speed / maximum speed for a longer time or drops to a lower speed / comes to rest	B1

10.

Question	Answer	Marks
7E(a)	magnetic field lines cut the solenoid / coils or changing magnetic field in solenoid	B1
	an electromotive force / e.m.f. is <u>induced</u>	B1
	electromotive force / e.m.f. produces the current	B1
7E(b)	negative / opposite / reversed current / deflection / reading	B1
	negative e.m.f. / voltage (induced) or magnetic field cut in opposite direction or magnetic field changes in opposite sense / decreases (owtte)	B1

11.

Question	Answer	Marks
7(a)	<u>current</u> (in coil) is (any one from) - away from A or towards B - from positive to negative - at right angles to magnetic field - makes the coil an (electro)magnet	B1
	left-hand rule stated, mentioned or used or force at right angle to field and current or top face of coil is a S-pole / bottom face a N-pole	B1
	<u>forces</u> in opposite directions (on opposite sides causing rotation) or unlike poles attract / like poles repel (only if coil is electromagnet)	B1
7(b)(i)	stronger magnetic field or (soft) iron becomes (temporary) magnet / magnetises easily	B1
	stronger / larger <u>force</u>	B1
7(b)(ii)	larger current / larger voltage / more turns on coil / wider coil / larger coil / thicker wire / stronger magnet	B1

12.

Question	Answer	Marks
6(a)	bar cuts <u>magnetic</u> field (lines / flux) or change in <u>magnetic</u> flux (in circuit with ammeter)	B1
	emf or current <u>induced</u>	B1
6(b)	negative (small)	B1
	no reading	B1
	no reading	B1
6(c)	move bar faster or wider magnet	B1
6(d)	force downwards (caused by current in magnetic field) or <u>force</u> opposes movement / motion	B1

13.

Question	Answer	Marks
9(a)	steel underlined and no other material indicated	B1
9(b)	use of (plotting) compass	B1
	place the (plotting) compass (on the paper next to the magnet) and mark a dot where the compass needle points	B1
	move compass / other end of needle next to the dot and mark a dot where the compass needle points	B1
	continue and needle points in the direction of the magnetic field	B1
9(c)(i)	(free) electrons attracted to K / positive terminal or repelled by J / negative terminal	B1
	(conventional current) from K to J or from positive or to negative or from right or to left	B1
9(c)(ii)	(Fleming) left-hand rule mentioned or equivalent rule mentioned	B1
	magnetic field related to index / pointer finger / equivalent	B1
	current related to middle finger / equivalent	B1
	thumb points down the page / equivalent and force down (the page)	B1
	or catapult field explanation: field lines clockwise when viewed from K or anticlockwise when viewed from J	B1
	extra field lines / stronger field above wire	B1
	fewer field lines / weaker field below wire	B1
	force towards weaker field	B1
9(c)(ii)	there is no force F or it disappears / becomes equal to zero	B1
	wire is magnetically screened / shielded by the iron tube	B1
9(d)	moment acts (on coil) or forces (on vertical sides) in opposite directions	B1
	currents (in vertical sides) or in opposite directions	B1
	or coil becomes / acts as a magnet or coil produces magnetic field	(B1)
	N face (of coil) attracted to S pole (of magnet) or S face (of coil) attracted to N pole (of magnet) or N face (of coil) repelled by N pole (of magnet) or S face (of coil) repelled by N pole (of magnet)	(B1)

14.

Question	Answer	Marks
6(a)	<u>iron</u> core and two coils wrapped around core	B1
	input and output clear and more turns on secondary / output coil	B1
6(b)	magnetic field mentioned	B1
	alternating current (a.c.) produces magnetic field that varies	B1
	core channels magnetic field to secondary coil and electromagnetic <u>induction</u>	B1
6(c)	less energy transferred to thermal energy in wires	B1
	(because) smaller current (for a given power)	B1

15.

Question	Answer	Marks
9(a)(i)	core becomes magnetised (due to current in coil)	B1
	iron armature attracted to core	B1
	contacts touch	B1
9(a)(ii)	steel is a permanent magnet or iron is a temporary magnet	B1
	with a steel core motor stays on or contacts stay closed (if switch S is opened)	B1
9(a)(iii)	(with relay) safer or switch S connected to low voltage or switch only handles low currents	B1
9(b)(i)	at least two lines inside coil and horizontal near centre and covering at least six wires of coil	B1
	correct shape of lines fanning out at ends of coil	B1
	at least two lines passing outside the coil from one end to the other	B1
	correct direction of field lines on at least one line and none wrong	B1
9(b)(ii)	N-pole on right hand side	B1
9(c)(i)	move magnet into / near coil or move coil	C1
	<u>move</u> magnet / coil <u>quickly</u> or <u>move</u> <u>strong</u> magnet	A1
9(c)(ii)	lines of magnetic field cut the coil or flux in coil changes	B1
	<u>induced</u> e.m.f. / voltage produced (in complete circuit)	B1

16.

Question	Answer	Marks
10(a)(i)	ammeter in series connecting switch with motor	B1
	voltmeter across motor or battery	B1
10(a)(ii)	(E =) VIt in any form numerical or algebraic or P = VI and P = E / t in any form	C1
	48 J	A1
10(a)(iii)	(E = mgh) in any form numerical or algebraic	C1
	$0.15 \times 0.8(0) \times 10$	C1
	1.2 J	A1
10(a)(iv)	ANY two of - motor / mass has kinetic energy - heat generated in motor / wires - heat generated in air due to drag / resistance <u>work done</u> to lift string	B2
10(b)(i)	mention of magnetic field (between N and S poles)	B1
	coil cuts lines of magnetic flux / magnetic field lines or flux / field in coil changes	B1
	induction	B1
10(b)(ii)	statement of Lenz's law , e.g. (induced) current flows in a direction to oppose the change producing it	B1
	coil becomes a magnet or current flowing in coil produces force in field	M1
	poles of coil attracted to opposite poles on magnet or force up on a side as side moves down	A1

17.

Question	Answer	Marks
6(a)(i)	4.0 V c.a.o.	B1
6(a)(ii)	(V =) $12 - 4.0$ or 8.0 or $4.0 / 700$ or 5.7×10^{-3}	C1
	(R =) V / I or $8.0 \times 1 / (4.0 / 700)$ or $700 \times (8.0 / 4.0)$ or $12 / (4.0 / 700)$ or 2100 (Ω)	C1
	1400 Ω	A1
6(b)	resistance of LDR decreases	B1
	current increases or p.d. across 700 Ω resistor / oscilloscope increases or p.d. across LDR decreases	M1
	trace moves up the screen	A1

18.

Question	Answer	Marks
7(a)(i)	current and <u>magnetic field</u> / <u>magnetic flux</u> (experiences a force) or the two <u>magnetic fields</u> interact	B1
7(a)(ii)	three separate approaches I / II / III	
7(a)(ii)I	B to A	B1
	left-hand rule mentioned / described	B1
	or	

7(a)(ii)II	B to A	B1
	top face of coil attracted to S-pole or top face acts as an N-pole	B1
	or	
7(a)(ii)III	B to A	B1
	more field lines under AB or fewer lines above AB	B1
7(b)	($I = $) Fx or $2 \times 9.6 \times 10^{-3} \times 0.025$ or $9.6 \times 10^{-3} \times 0.025$ or 2.4×10^{-4} (N m) or 0.024 (N cm)	C1
	4.8×10^{-4} N m or 0.048 N cm	A1
7(c)(i)	split-ring commutator or (carbon) brushes	B1
7(c)(ii)	(soft-)iron core or stronger magnet(s) or larger current or more turns or smaller resistance or greater electromotive force (e.m.f.)	B1

19.

Question	Answer	Marks
8(a)	<u>surround</u> the component with a box or use of iron mentioned	B1
	<u>surround the component with (a box of) iron</u>	B1
8(b)(i)	solenoid and d.c. power supply and switch	B1
	core	B1
	<u>iron</u> core	B1
8(b)(ii)	named or described use	B1
	what happens when electromagnet is switched on	B1
	<u>current</u> causes magnetisation or description of what happens when electromagnet is switched off	B1
8(c)(i)	ammeters deflects / shows a reading	B1
	magnetic field lines cut by solenoid or changing magnetic field in coil	B1
	e.m.f. / voltage <u>induced</u>	B1
8(c)(ii)	no deflection and no (magnetic) field lines cut or constant field	B1
8(c)(iii)	larg(er) deflection	B1
	opposite deflection	B1
	field lines cut in opposite sense or opposite change in (magnetic) field or field lines cut faster	B1

20.

Question	Answer	Marks
7(a)	current direction in copper wire correct on any part of diagram	B1
7(b)	current and <u>magnetic</u> field create a force or current (in wire) creates a <u>magnetic</u> field	B1
	left-hand rule mentioned or used or interaction of magnetic fields of magnet and current (catapult effect)	B1
7(c)	(electric) motor or any device that contains a motor	B1
	(loud)speaker or any device that contains a loudspeaker	B1

21.

Question	Answer	Marks
7(a)(i)	<u>magnetic</u> field / flux mentioned	B1
	change in field / flux (in coil) or field (lines) cut coil / wire	B1
7(a)(ii)	more turns or stronger magnet or move magnet faster	B1
7(b)(i)	 correct symbol	B1
7(b)(ii)	converts more of the <u>input energy / power</u> into light / useful energy / power or less energy / power wasted <u>with same input energy / power</u> or <u>in same time</u> or <u>same</u> input power / energy / current gives more light / less energy wasted / power wasted / heat / greater power out or <u>less</u> input power / energy gives same output power / energy	B1

22.

Question	Answer	Marks
7(a)	magnetic (material)	B1
	temporary / soft magnetic (material)	B1
7(b)(i)	it / a.c. changes direction or changes polarity / from positive to negative (continually)	B1
	it / a.c. has varying size or is sinusoidal / like a sine wave	B1
7(b)(ii)	magnetic field (in core / secondary coil) is not changing / remains constant	B1
	no (electromagnetic) <u>induction</u>	B1

23.

Question	Answer	Marks
7(a)	magnet is attracted to magnetic material or is not attracted to the non-magnetic material	B1
7(b)(i)	(direction) to the left or away <u>from region 2</u>	B1
	like poles / N-poles repel	B1
7(b)(ii)	correct direction on at least one of the eight lines and none wrong anywhere else	B1
7(b)(iii)	1 field (lines) cut the coil or field/flux (in coil) changes	B1
	2 voltage large(r) or magnetic field is strong(er) / to be in the (magnetic) field or more field (lines) cut (in a given time) / so field lines are cut or field lines close(r) together / field lines cut in small(er) time	B1

24.

Question	Answer	Marks
11(a)(i)	steel	B1 B1
11(a)(ii)	iron	
11(a)(iii)	iron B1 (i) and (iii) both correct B1 (ii) correct	
11(b)(i)	clear use (e.g. change voltage / current)	M1
	detail of operation (to transmit electricity or in a charger etc.)	A1
11(b)(ii)	1 voltage on vertical axis and time on horizontal axis	B1
	clear attempt at sinusoidal curve for at least one cycle	B1
	2 ($V_s =$) $V_p N_s / N_p$ or $220 \times 85 / 1700$	C1
	11 V	A1
11(c)(i)	1 magnetic field / flux / flux linkage mentioned	B1
	magnetic field (lines) cut solenoid or v.v. or changing (magnetic) field / flux / flux linkage (in solenoid)	B1
	induced e.m.f. / voltage	B1
	2 it / current magnetises solenoid / produces magnetic field / flux(linkage) in solenoid / <u>produces</u> a S-pole (in solenoid)	B1
	magnet repelled or experiences a force to the left	B1
11(c)(ii)	smaller reading / deflection	B1
	reading in opposite direction	B1

25.

Question	Answer	Marks
7(a)	PQ: a force towards the top of the page (second box) ticked	B1
	QR: a force towards the right of the page (bottom box) ticked	B1
7(b)	current reversed	B1
	both forces reversed	B1
7(c)	force(s) decrease or less heat generated	B1

26.

Question	Answer	Marks
7(a)(i)	current in coil (at right angles) in a magnetic field (of magnet) or left-hand rule mentioned	B1
7(a)(ii)	reverses / changes direction of current (in coil)	B1
	reverses current every half turn / when coil is vertical or reverses forces (on side AB / CD) or keeps forces in same direction for wire on one side	B1
7(b)(i)	($E =$) $\mathcal{V}It$ or $2 \times 12 \times 8$ or $E = Pt$ and $P = VI$ or $E = VQ$ and $Q = It$	C1
	190 or 192 J	A1
7(b)(ii)	73% or 0.73	B1

27.

Question	Answer	Marks
7(a)(i)	arrow or other indication left to right in rod or correct in circuit labelled C	B1
7(a)(ii)	arrow from N-pole to S-pole labelled M	B1
7(a)(iii)	arrow downwards labelled F	B1
7(b)(i)	rod cuts magnetic field / flux lines or flux (in circuit) changes	B1
	induced e.m.f. / voltage / current formed	B1
7(b)(ii)	move rod faster	B1

28.

- (a) (i) X N-pole B1
Y S-pole **and** Z N-pole B1
- (ii) they touch / move towards each other **and** opposite poles attract B1
- (b) any sensible use: starting-motor circuit; with a logic gate; B1
nuclear power station
corresponding explanation: current too large for dash-board switch; B1
current too small to power device; too dangerous to reach switch

29.

- (a) current / a.c (in primary coil) creates magnetic field B1
or current / a.c magnetises iron
changing magnetic field (in secondary) B1
- (b) it / secondary has less turns (than primary) B1
or primary has more turns (than secondary)
or (some) flux escapes
- (c) (steel is) a permanent magnet B1
or weaker fields produced
or (steel) difficult to magnetise / demagnetise
or (steel) is a hard magnetic material
- (d) passes current / charge in one direction B1
or has high resistance / is an insulator when current in
one direction / reverse biased

30.		
(a)	(current in coil) creates magnetic field or current is at right angles to magnetic field (of permanent/cylindrical magnets)	C1
(b)	into and out of magnet or left and right or backwards and forwards	B1
	current is one way then reverses (so reverses force)	B1
(c)	$(\lambda =) v/f$ numerically or algebraic in any form 0.64 m	C1 A1
31.		
(a)	(i) diagram showing coil of wire and either • magnet or • another coil and supply (dc and switch or ac) coil of wire connected to an ammeter or voltmeter or cro or other method of detection, e.g. lamp magnet or coil moved or <u>change in current</u> mentioned if another coil used	B1 B1 B1
	(ii) ANY 2 from • move magnet (or coil) faster • larger current in primary (if transformer drawn) • more turns in coil • stronger magnet (if magnet drawn) • soft iron core	B2
	(iii) 1 direction of <u>induced</u> current/ <u>induced</u> emf opposes the change (that produces it)	B1 B1
	(iii) 2 (magnetic) flux/field/poles in coil caused by movement/(induced) current in coil statement of how opposition occurs, e.g. repulsion as magnet moves in; N pole created (by induction) at end of coil as N pole approaches	B1 B1
(b)	(i) 1 $(I =) P/V$ numerical or algebraic 15(.15) A	C1 A1
	2 $(E =) Pt$ or VIt or $500(000) \times 60 \times 60$ $1.8 \times 10^9 \text{ J}$ or 500 kWh	C1 A1
	(ii) low current $P = I^2R$ or $E = I^2Rt$ explained	B1 B1

32.

- (a) (i) no free electrons (in plastic) **or** all electrons are bound / structural B1
- (ii) (aluminium) is not magnet(ic) **or** cannot be magnetised B1
(iron) is a temporary / soft magnetic material **or** is not a permanent magnet B1
- (b) (i) magnetic field / flux (mentioned) B1
(magnetic) field lines cut wire / solenoid / circuit **or** changing magnetic field / flux B1
voltage/e.m.f. **induced** B1
- (ii) ($V =)IR$ **or** 0.045×1.2 **or** 0.000045×1.2 C1
 $5.4 \times 10^{-5} \text{ V}$ **or** 0.054 mV A1
($Q =)It$ **or** 0.045×0.14 **or** 0.000045×0.14 C1
 $6.3 \times 10^{-6} \text{ C}$ **or** 0.0063 mC A1
- (c) (i) larger **or** twice the current B1
(magnetic) field lines cut faster **or** (magnetic) field changes faster **or** twice the current B1
- (ii) double the current for half the time **or** larger current for less time **or** product $I \times t$ is the same B1
- (d) any **two** from:
insert S-pole (at same end)
insert (N-pole) at other end **or** from other direction
withdraw N-pole (from same end implied)
withdraw S-pole from other end **or** pass through completely B2

33.

- (a) (i) at least two straight parallel lines inside the coil B1
 at least two (complete) lines one above the coil and one below the coil B1
 third line in middle **and** evenly spaced **and** two closed loops B1
 (any crossings max. 2/3)
- (ii) current (in X) increases B1
 magnetic field becomes stronger / changes B1
 current / e.m.f. / voltage **induced** in Y / electromagnetic induction B1
 opposite deflection B1
 larger deflection B1
 magnetic field decreasing **or** quicker (rate of) change B1
- (b) (i) to increase the strength of the magnetic field B1
 to direct / concentrate the magnetic field (into the secondary coil) B1
- (ii) $(P =)VI$ **or** $33\,000 \times 85$ C1
 $2.8 \times 10^6 \text{ W}$ **or** 2800 kW **or** 2.8 MW A1
- (iii) $(E =)VIt$ **or** $33\,000 \times 85 \times 3600$ **or** $2.8 \times 10^6 \times 3600$ C1
 $1.0 / 1.01 / 1.008 \times 10^{10} \text{ J}$ A1

34.

- (a) (i) mention of (magnetic) field / flux (of N and S-poles) C1
 (coil/wire) cuts **magnetic** field / flux / lines A1
or magnetic flux in coil changes
- (ii) (one side of) coil cuts one way and then the other B1
or (side) moves one way and then the other / returns
or flux increases and then decreases
- (b) increase in emf for both stronger magnets and more turns B1
 no change / same frequency for both stronger magnets and more turns B1
 increase and increase for turn the coil faster B1

- 35.
- (a) (i) 1 S-pole on right of core B1
- 2 N-pole anywhere on vertical section of armature B1
and S-pole anywhere on horizontal section of armature
or
 N-pole on left of vertical section of armature **and** S-pole on right
- (ii) poles (on core) reverse / change positions B1
 (armature still) attracted (to core) B1
- (iii) (iron is a) temporary magnet B1
or (iron) easily demagnetised
or steel retains magnetism
 when current off / no battery / switch off / circuit open B1
and
 armature released / does not stay attracted / opens connections (at AB)
- (b) (i) thermistor B1
- (ii) resistance (of X) decreases B1
 current (in coil) increases **or** more voltage across coil B1
and
either relay switch closes **or** circuit (to bell) complete
- (iii) 1 $(V=) IR$ **or** $1.5 (\times 10^{-3}) \times 2000$ C1
 $3(.0)V$ A1
- 2 $9(.0)V$ B1
- 3 $12/200$ **or** $0.06 (A)$ **or** $60 (mA)$ seen C1
or $(R_T =) 195(.12 \Omega)$
 $61(.5)mA$ **or** $0.061(5)A$ **or** $62 mA$ **or** $0.062 A$ A1
- (iv) light dependent resistor **or** LDR B1
- 36.
- (a) field lines **of magnet** mentioned or magnetic flux mentioned B1
 field lines cut the coil or flux changes B1
- (b) reversed movement of magnet causes one of B1
- reversal of (induced) emf
 - reversal of (induced) current
 - field lines cut / flux change in reverse direction
- LED emits light when **current** passes in one direction B1
- (c) more current or more induced emf B1
and flux lines cut faster or faster change in flux

- 37.
- (a) one label correct and not contradicted C1
C, 1S and 1B all correct and clear and none contradicted A1
- (b) any **three** from:
magnetic field (between poles)
(coil/wire) cuts field/flux **or** field/flux cuts (coil/wire) **or** field/flux changes
(electromagnetic) induction
brushes rub against/in contact with rings B3
- (c) (half) distance across screen **or** count divisions of/measure wavelength **or** the
wavelength corresponds to one rotation B1
half distance multiplied by time base setting B1
- 38.
- (a) wire cuts field lines B1
current/e.m.f./voltage **induced** B1
- (b) larger deflection **and** to the left/opposite direction B1
- (c) no deflection/current B1
- 39.
- (a) arrows on long sides in opposite vertical directions [B1]
downwards on right **and** upwards on left **or** correct rotation shown [B1]
- (b) no (horizontal) distance between forces **or** forces through axle/pivot/axis [B1]
- (c) two halves of split ring clear and clearly connected to each end of coil [B1]
contacts/brushes labelled or described **and** connected to battery [B1]
each side of split ring touches other terminal/brush **or** current reverses in coil
or changes terminals of connection to battery [B1]
forces reverse on **sides** of coil **or** forces always in same direction on **side**
nearest a pole [B1]

- 40.
- (a) oval/circle through or near A centered on or near X
clockwise arrow on line(s) around X **and** none wrong B1
B1
- (b) fields (due to X and Y) cancel **or** X and Y fields equal and opposite B1
- (c) (i) to the left
or towards X/A/B B1
- (ii) current (in wire Y) and (magnetic) field (caused by other wire)
or two (magnetic) fields interact B1
- 41.
- (a) steel / magnadur / alnico / magnetite B1
- (b) (i) mention of cutting (lines of) magnetic field / change in (magnetic) flux M1
great(est) rate of change A1
or fast(est) cutting
or other explanation involving time
- (ii) vertical/upright B1
or turned through 90°
or normal to (magnetic) field
- 42.
- (a) (soft) iron/mu-metal B1
- (b) magnetic field **or** flux **or** flux/magnetic lines mentioned B1
changing magnetic field **or** changing flux **or** flux lines cut coil B1
induced voltage/current/e.m.f. B1
- (c) less power/energy/heat loss (**allow** no power loss/to prevent power loss) **or**
more efficient **or** thinner wire can be used (**ign.** cheaper) B1

- 43.
- (a) (i) $4.5 + 0.3$ **or** 4.8 C1
 $(I =)V/R$ **or** $12/4.8$ **or** $12/4.5$ **or** $12/0.3$ **or** $12/0.28125$ C1
2.5 A A1
- (ii) decrease resistance (of variable resistor) B1
increase current (in solenoid) B1
- (iii) 1. force on PQ/wire **or** PQ/wire moves M1
force/movement out of page/outwards/towards observer
(not upwards) A1
2. force/speed/acceleration larger B1
- (b) (i) $(P =)VI$ **or** 75×12 C1
900 W A1
- (ii) (thick wires) have low resistance B1
(thick wires) not as hot/do not melt B1
- (c) current to relay/coil/solenoid/electromagnet B1
core/relay/coil/solenoid/electromagnet magnetised B1
connections made (in motor circuit) B1
- 44.
- (a) (at least) **two** parallel horizontal lines within the cylinder B1
(at least) **two** correctly shaped lines outside the cylinder B1
- (b) (i)  (right to left) **and** on diagram (somewhere) B1
- (ii) 1. path continuously curving in same direction M1
upwards (**ignore** lines outside the shaded area) A1
2. (changes to) downwards (curve) **not** reverses/opposite direction B1
- 45.
- (a) (i) conventional current direction correct in coil/one lead B1
- (ii) at least 1 line axially through coil A
OR line above and below end of coil A B1
- at least two curved lines in ring from ends of A
to ends of B (and inside A and B) B1
- correct direction on at least one line/arrow for candidate's (i) B1

(b) (i)	(magnetic) flux/field cuts (coil B) OR field/flux changes (in coil B)	B1
	induces an e.m.f./voltage/current (in B)	B1
(ii)	(voltmeter) deflects to left/opposite (and returns to zero)	B1
	flux/field decreases/collapses/reduces OR iron loses magnetism OR change in field is in opposite direction OR to oppose flux/field change	B1
(iii)	ANY 2 lines more turns on coil B larger voltage/current (e.g. of battery)/more cells battery smaller (internal) resistance smaller resistance of wires; thicker wires; shorter wires thicker or shorter iron ring; use soft iron; coil A and B closer (on ring); more sensitive voltmeter; laminate the iron ring	B2
(c) (i)	$(P =) VI$ algebraic or numerical 384 W OR 380 W	C1 A1
(ii)	$(P =) I^2 R$ OR $(P =) V^2/R$ OR VI and V/R seen algebraic or numerical OR clear voltage of 4(.0 V) or 8(.0 V) seen	C1
	$1.6^2 \times 2.5$ OR $1.6^2 \times 5$ OR (power) 6.4 (W) seen	C1
	12.8 W OR 13 W	A1

46.		
(a)	(i) iron; soft iron; mu-metal	B1
	(ii) rod becomes (an induced) magnet opposite poles attract; N attracts S OR magnetic pole(s) on rod/at P reverses (due to induced magnetism)	B1 B1
(b)	(i) at least two circles centred on wire (no crossings) clockwise arrow on at least one circle and no arrows wrong	B1 B1
	(ii) lines closer together	B1
47.		
(a)	(i) N at top end of bar and S at bottom end	B1
	(ii) attracted to/moves towards iron core unlike poles attract	B1 B1
(b)	they disappear/bar is demagnetised/loses its poles/is weaker	B1
48.		
(a)	(i) recognisable sine/cosine curve (≥ 2.0 cycles)	B1
	(ii) larger (peak)(voltage) higher frequency/shorter period/described in words (allow shorter wavelength)	B1 B1
(b)	$(R =) V/I$ or $12/0.50$ $24\ \Omega$	C1 A1

49.

(a) upward arrow (**not** curved) on iron bar B1

(b) attraction/force not enough//weight of bar too high//friction at pivot/with copper bar B1

(c) **three** of:

electromagnet works // magnetic field created
iron bar moves/lifted up
spring pulls copper bar across/contracts
contacts break circuit // contacts open

B3

(d) more turns in coil // more iron in electromagnet // electromagnet nearer iron bar //
iron bar less weight // weaker spring
correct explanation which involves force on iron bar

M1

A1

50.

(a) all three correct: force, field, current B1

(b) (i) Fd // 20×4 ; 20×2 ; 20×0.04 ; 20×0.02 ; $20 \times 4 \times 2$; 160
80 N cm, 0.8 N m

C1

A1

(ii) more turns (on coil)
use soft iron
more current
increase AB or CD
increase BC or AD ANY 2

B1

51.

- (a) (i) d.c. current flows in one direction **or**
a.c. current flows in one direction then the other B1
- (ii) mention of magnetic field/flux B1
(magnetic) field lines // flux cuts coil // flux changes in coil B1
induction of voltage/current B1
something **relevant** reverses (e.g. field/flux cuts in one direction then the other // N pole approaches then leaves // N pole approaches and S pole approaches)
and link to a.c. B1
- (iii) **two** of: thicker wires; more turns of coil; stronger magnet; faster rotation; lower resistance (of lamp) B2
- (b) (i) to reduce heat/energy/power loss (on the power lines) B1
(higher voltage means) lower current B1
- (ii) 25:400 // 1:16 // 0.0625 B1
- (iii) reduces resistance // less power loss // costs less to run // more current // more power B1
increases weight // more support needed // more wind resistance // more ice forms // costs more to install B1
- (iv) 13 A B1
($I = P/V$) // 4.2 A // 4.17 etc A B1
must choose higher value to avoid fuse blowing // other fuses melt B1

52.

- (a) remain stationary/no effect/unaffected B1
- (b) lifted up/attracted/stick to rod (stated not implied) B1
fall down/return to dish B1
- (c) lifted up/attracted/stick to rod (stated not implied) B1
stay up/remain attracted B1

- 53.
- (a) (i) current in magnetic field **or** motor effect/LH rule **or** coil is magnet (produces) force **or** current **direction** changes **or** coil moves **or** repulsion **and** attraction
force changes direction/backwards and forwards
- (ii) air (atoms/molecules/particles) (next to cone) vibrates
compressions **and** rarefactions **or** high **and** low pressure
or vibrations passed on **or** longitudinal
- (b) the note is louder/has greater intensity (**not** changed frequency)
- 54.
- (a) both arrows point inwards
both arrows extended must pass through base of bar magnet
N marked on both needles nearest S pole
- (b) (current causes) coil/iron to become magnet/an electromagnet/creates magnetic field
iron (in coil) attracts/pulls (pivoted) iron/armature
(and L-shaped iron rotates/moves/turns) not contacts attract each other
- (c) (i) resistance decreases
- (ii) series circuit/loop with C and lamp
completely correct circuit with a battery
- 55.
- (a) meter deflects (one way)
changing magnetic field/flux in ring/coil or cutting of flux/field
induces voltage/current
- (b) ammeter returns to/remains at zero
- (c) ammeter deflects in opposite direction (then returns to zero)
field decreases/change in opposite direction **or** field/flux cuts in opposite direction

56.		
(a)	NS marked on each piece correctly	B1
(b)	NS/unlike/opposite poles attract	B1
	switch closes or soft-iron/contacts touch	B1
(c)	(i) resistance decreases	B1
	(ii) current increases clearly in coil/through thermistor	B1
	magnetic field (in coil) (and contacts close)	B1
57.		
(a)	(i) from N to S or towards right	B1
	(ii) downwards	B1
(b)	(i) rough circle around each wire (–1 any crossing lines)	B1
	correct shape around both wires or large circle around both wires	B1
	direction of field correct on any one correct line and no direction wrong	B1
	(ii) attractive force drawn on/near each wire	B1
58.		
(a)	(i) steel	B1
	(ii) rod inside (coil) with current on (at some stage)	B1
(b)	(i) (soft) iron accept Mumetal or any other soft magnetic material	B1
	(ii) all lines directly join from left to right and top line goes down and bottom line up	M1
	no lines inside box and no lines cross/touch	A1
59.		
(a)	at least 3 approx straight lines in middle of coil	B1
	one loop around top or bottom of coil	B1
	correct direction shown on at least one line or loop	B1
(b)	(i) 100 oscillations/cycles/waves/changes in direction (and back again) in 1 sec	
	or time for one oscillation etc. 0.01s	B1
	(ii) coil/cone moves left to right or back and forth or oscillates/vibrates or in and out	B1
	(current in) coil produces magnetic field/pole(s)	
	or Left Hand Rule/current in magnetic field (gives force)	B1
	fields/poles (of coil) oscillate/reverse	B1
	force (exerted by magnet) oscillates/alternates (accept attracts/repels)	B1 ANY 2 B2

60.			
(a)	electromagnetic/em induction or induced current/e.m.f. (NOT magnetic/electric induction)		B1
(b)	deflects to left/opposite deflection		B1
(c)	nothing or no deflection/current/e.m.f. or needle stationary no lines of flux are cut or no change in magnetic field		B1 B1
61.			
(a)	any mention of magnetic field coil cuts lines of (magnetic) flux/field lines		C1 A1
(b)	pass current / connect coil to output / prevent wires tangling		B1
(c)	(induced) voltage or current opposes the change (producing it)		B1
(d)	large(r) coil, strong(er) magnet, iron inside coil, more turns (on coil)		B1
62.			
(a)	Y input and ground connected across resistor		B1
(b)	3 squares or 3 x 2 6V		C1 A1
(c)	line drawn at 1.5 squares		B1
63.			
(a)	rods magnetised with like poles next to each other e.g. both rods N at one end like poles repel		B1 B1
(b)	nothing happens copper is not magnetic	accept eddy currents (induced) changing flux induces current/voltage	B1 B1
64.			
(a)	arrows in A and C to right arrow in B to left or right if both A and C to left		B1 B1
(b) (i)	SNSN or NSNS		B1
(ii)	they/iron pieces attract/move together attraction of opposite poles/unlike poles/S and N	e.c.f. (i) throughout	B1 B1
(c) (i)	opposite direction/reverses/poles change		B1
(ii)	weaker (field) or (iron) demagnetises		B1

65.

(a) to step down/reduce the **voltage**

(**ignore** just step down transformer)

B1

(b) two coils (no label needed)

(**not** if primary connects secondary)

B1

output/secondary has fewer turns than input/primary clear; coils labelled

or right-hand coil has fewer turns

B1

complete (soft) iron (core) **labelled**

(**ignore** circuit symbol)

B1

(c) (i) less energy/power/heat loss/heating

(**ignore** just more efficient)

B1

current is reduced/low

(**not** if resistance changes)

B1

(ii) resistance is decrease

(resitivity is not resistance)

B1

electrical power/energy related to resistance

e.g. $P = I^2R$, P prop to R (**not** V^2/R alone)

or resistance $\propto 1/\text{area}$

(**accept** power related to R etc. given in (i))

B1

66.

(a) (i) magnetic (field) of current / coil / recording head

or head is magnetized / an electromagnet

B1

(ii) magnetism / magnetic field or current or poles on head reverses / changes direction

(**accept** "due to alternating current")

B1

(iii) each direction / one cycle longer (on tape)

B1

(b) (i) need to keep record / tape stored or played

B1

(ii) iron, steel etc

B1